

Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

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Education

University of Wyoming, Energy & Petroleum Engineering

Laramie, WY, USA
2025 – present

- Advisor: Dr. Soheil Saraji
- Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam
- Coursework: Rock Mechanics, Enhanced Oil Recovery, Graduate English Writing, Geostatistics, Python for Engineers

Pusan National University, Naval Architecture & Ocean Engineering

South Korea
2018 – 2024

- Advisor: Dr. Inwon Lee
- Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD

Research Experience

University of Wyoming, Graduate Research Assistant

Laramie, WY, USA
2025 – present

- Developing a coupled solid–fluid simulation framework in solids4Foam (OpenFOAM-based) to model proppant grain behavior under compressive loading between parallel rock fracture walls
- Implementing J2 perfect-plasticity rock constitutive models and penalty-based contact algorithms to capture proppant embedment and fracture wall deformation
- Constructing structured O-grid meshes for proppant–rock contact problems; validated stress distributions against Hertz contact theory
- Performing laboratory training on triaxial cell operation, confining/pore-pressure control, and rock compressibility (bulk modulus) testing using DCI equipment

KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time)

South Korea
2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Investigated gravity–capillary wave dynamics on elastic silicone films subjected to continuous air-blow loading above water
- Derived dispersion relations for gas–solid–liquid interface interactions under nonlinear elastic coupling
- Developed 2D ventilated supercavitation models and introduced wake-theory-based modifications, significantly improving agreement between theoretical predictions and experimental data for cavitator-driven cavity asymmetry
- Conducted vortex–wave interaction CFD simulations near free surfaces using ANSYS Fluent; analyzed free-surface-induced von Kármán vortex asymmetry mechanisms

Pusan National University — Department of Naval Architecture & Ocean Engineering, Undergraduate Research Assistant

South Korea
2022 – 2024

- Performed RANS CFD validation of KCS hull resistance (Gothenburg 2010 Workshop benchmark) using STAR-CCM+; 1.16% error in total resistance coefficient CT ; contributions: structured mesh optimization and post-processing
- Performed particle image velocimetry (PIV), hull resistance testing, and propulsion system design per ITTC 1957/1978 standards
- Developed ANN-based defect detection models using vibration and acoustic sensor data for ship structural health monitoring (MATLAB & Python)

Industry Experience

Samsung Heavy Industries Laboratory , Mechanical Engineer Intern	South Korea
Trained and evaluated artificial neural network architectures for Green Nuri vessel defect classification using MATLAB and Python	2024 – 2024
Pusan National University , Capstone Design Project	South Korea
- Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria - Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability	2023 – 2023

Publications

Computational analysis of proppant embedment and fracture conductivity in deformable rock using solids4Foam	in preparation
Seungmin Yome, Soheil Saraji (Computers and Geotechnics)	
Numerical Recovery of Hydroelastic Wave Dispersion in Coupled Elastic Film–Liquid Systems Using a Finite-Volume Fluid–Structure Interaction Solver	in preparation
Seungmin Yome (Journal of Fluids and Structures)	

Grants & Fellowships

Graduate Research Assistantship (fully funded)	2025–Present
University of Wyoming	
Research Scholarship	Jan 2024
KAIST	
Undergraduate Research Grants	2022–2024
Pusan National University	
Academic Excellence Scholarship (2nd award)	Jan 2025
Pusan National University	
Academic Excellence Scholarship (1st award)	Jan 2024
Pusan National University	
Foreign Language Special Lecture Consignment Program Scholarship	Jan 2021
Pusan National University	
Creative Human Resources Development Project Group Scholarship	Jan 2018
Pusan National University	

Selected Activities & Awards

Invited Participant — Samsung Heavy Industries Wind Propulsion Ship Forum (travel award)	Jan 2024
Samsung Heavy Industries	
Invited Speaker — "Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship"	Jan 2024
Korean Society of Shipbuilding & Marine Engineering / Coast Guard	

Invited Participant — Hyundai Heavy Industries Eco-Friendly Smart Ship Forum (travel award)

Jan 2023

- Hyundai Heavy Industries

TPC (Test Preparation Course) Certification — English Consignment Program

Jan 2021

- Pusan National University

Teaching & Tutoring

Private Tutoring (Educational Volunteering), Mathematics Instructor

South Korea

- Taught high school mathematics to undergraduate-level students as part of educational volunteering activities over two years

2022 – 2024

Private Tutoring, Physics & Chemistry Instructor — College Entrance Preparation

South Korea

- Taught physics and chemistry to 10 high school seniors (Grade 12) preparing for the Korean college entrance exam (CSAT) over a 6-month intensive period
- 9 out of 10 students (90%+) gained admission to top-tier Korean universities

2023 – 2024

Skills

Simulation & CAE

Programming

Design & Engineering Tools

Laboratory

Languages

Korean

Native

English

Fluent